

Barbara Saget

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Education

Paris School of Economics, Université Paris I Panthéon-Sorbonne PhD in Economics ★ Supervisors: Mouez Fodha and Katheline Schubert. ★ Fields: Environmental economics, dynamic modeling. ★ Subject: "Essays on carbon neutrality policy options: negative emissions and circular economy". ★ Research topics: Net-zero emissions, carbon dioxide removal, energy transition, circular economy.	<i>September 2024 -</i>
Paris School of Economics, Université Paris I Panthéon-Sorbonne Master's degree Analysis and Policy in Economics ★ Highest honours. ★ Master's thesis in environmental economics under the supervision of Katheline Schubert.	<i>September 2021 - June 2024</i>
Université Paris I Panthéon-Sorbonne Bachelor's degree « Magistère d'Economie » ★ High honours.	<i>September 2020 - June 2021</i>
Lycée Saint-Marie-de-Neuilly Preparatory Class in Social Sciences (CPGE B/L) ★ Admitted to the Ecole Normale Supérieure de Lyon.	<i>September 2017 - June 2020</i>

Professional experience

OECD, Center for Tax Policy and Administration Intern in the Tax Policy and Statistics Division, "Personal and property tax" unit.	<i>February 2023 - August 2023</i>
French Treasury Intern in the Economic Department of the French embassy at Prague.	<i>August 2022 - January 2023</i>
Hospinnomics (Research chair in partnership between PSE and the AP-HP) Research assistant for Professor Lise Rochaix. Health Economics.	<i>May 2021 - July 2021</i>

Teaching

Université Paris 1 Panthéon-Sorbonne Teaching assistant - <i>Microeconomics</i> , Stéphane Gauthier (L2). Competitive equilibrium and market failures.	<i>September 2024 - January 2026</i>
Sciences Po Paris Teaching assistant - <i>Economics</i> , Isabelle Méjean (L1). Introduction to economics.	<i>February 2025 - May 2025</i>
Université Paris-Cité Teaching assistant - <i>Macroeconomics</i> , Olivier Allain (L1). Classical and Keynesian macroeconomics.	<i>February 2023 - May 2023</i>

Skills and interests

Langage	French (native), English (advanced).
Computer	R, Matlab, Python, Julia, LATEX, Excel.
Music	• September 2020 – today : Violinist in the Paris Sciences et Lettres (PSL) orchestra. • 2005-2017 : Student in the Conservatoire Régional of Boulogne-Billancourt (Music School Diploma).

Research

Working paper

- Saget, Barbara & Schubert, Katheline (2025). *CDR: opportunity or threat in the path to net-zero?*

Abstract: The European Union's commitment to net-zero carbon emissions by 2050 presents a significant challenge, particularly in hard-to-abate sectors such as aviation, shipping, and construction. This paper develops a dynamic model to characterize the optimal transition pathways to net-zero, integrating both mitigation and carbon dioxide removal (CDR) strategies. We distinguish between nature-based CDR, which provides only temporary carbon storage, and technological CDR, which offers permanent sequestration but remains costly and technologically immature. We incorporate three key constraints: (1) a carbon budget limiting cumulative net CO₂ emissions to meet climate targets, (2) hard-to-abate emissions from sectors lacking viable decarbonization options, and (3) limited land availability for nature-based CDR. We show that while both CDR methods facilitate a smoother transition to net-zero emission in the medium run, technological CDR may crowd out abatement efforts in the long run by offsetting both unavoidable and abatable emissions. Furthermore, reliance on nature-based CDR should decline over time as carbon leakage from temporary carbon sinks drives up the social cost of carbon until the steady state is reached. These findings highlight the crucial role of CDR in achieving net-zero emissions while emphasizing the need for policies that prevent mitigation deterrence and the overuse of temporary nature-based carbon storage.

Research projects

- Technological uncertainty and myopic behavior in the future reliability of CDR for achieving net-zero targets: under- or overconfidence in cost reductions from learning-by-doing.
- Assessing the real value of carbon offsets in voluntary carbon markets, with a focus on leakage issues (e.g. delocalization of deforestation projects to unprotected areas).
- Circular economy in hard-to-abate sectors: designing optimal recycling policies for the cement sector.

Conferences, seminars and summer school

Conferences

- FAERE Doctoral workshop - Annecy, France (May 2025)
- EAERE Annual Conference - Bergen, Norway (June 2025)
- LAGV Conference - Marseille (July 2025)
- FAERE Annual Conference - Nantes, France (September 2025)
- CentraleSupélec Working Paper Days - Paris, France (September 2025)

Seminars

- Regulation and Environment Seminar PSE (Paris, France)
- Macroeconomics Seminar PSE (Paris, France)

Summer school

- FERS School on carbon dioxide removal - Bertinoro, Italy (June 2025)